9. Demonstrate the Recurrent Neural Networks, algorithm by taking suitable data for any simple Application.

```
import tensorflow as tf
from tensorflow.keras.datasets import imdb
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import SimpleRNN, Dense, Embedding

# Load dataset
vocab_size = 10000
(X_train, y_train), (X_test, y_test) =
imdb.load_data(num_words=vocab_size)

# Preprocess (pad sequences)
max_length = 200
X_train = pad_sequences(X_train, maxlen=max_length)
X_test = pad_sequences(X_test, maxlen=max_length)

# Build RNN model
model = Sequential()

# Embedding layer
model.add(Embedding(input_dim=vocab_size, output_dim=32,
input_length=max_length))

# RNN layer
model.add(SimpleRNN(32))

# Output layer
model.add(Dense(1, activation='sigmoid'))

# Compile model
model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])

# Train model
model.fit(X_train, y_train, epochs=5, batch_size=64)

# Evaluate
loss, accuracy = model.evaluate(X_test, y_test)
print("Test Accuracy:", accuracy)
```